

15th September 2026, Linear Algebra Notes,
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Covered: Inner product space isomorphisms; norm and metric isomorphisms; polarization; best approximation in finite-dimensional IPVS; existence and uniqueness for closed convex sets; orthogonality criterion for subspaces.

1 Isomorphisms over Inner Products

Let $\{u_1, \dots, u_n\}$ be an orthonormal basis of V . For $x \in V$, there exist $\alpha_1, \dots, \alpha_n \in \mathbb{F}$ such that $x = \sum_{i=1}^n \alpha_i u_i$. Since $\langle x, u_j \rangle = \alpha_j$, we get $x = \sum_{i=1}^n \langle x, u_i \rangle u_i$.

Consider the linear map $T : V \rightarrow \mathbb{F}^n$ as $x \mapsto (\langle x, u_1 \rangle, \langle x, u_2 \rangle, \dots, \langle x, u_n \rangle)$.

$$\begin{aligned}
 \text{Let } w \in \ker T &\implies Tw = 0 \\
 &\iff \langle w, u_i \rangle = 0 \quad (1 \leq i \leq n) \\
 &\iff \langle v, w \rangle = 0 \quad \forall v \in V \\
 &\iff \langle w, w \rangle = 0 \\
 &\iff w = 0 \\
 &\implies \ker T = \{0\}.
 \end{aligned}$$

Note, that $\dim V = \dim \mathbb{F}^n = n$ and T is an injective map which implies T is an isomorphism. V and $\mathbb{F}^n$ are thus isomorphic as vector spaces.

$$\begin{aligned}
 \|T(x)\|^2 &= \langle T(x), T(x) \rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2 \\
 \|x\|^2 &= \langle x, x \rangle = \left\langle \sum_{i=1}^n \langle x, u_i \rangle u_i, \sum_{j=1}^n \langle x, u_j \rangle u_j \right\rangle = \sum_{i=1}^n |\langle x, u_i \rangle|^2
 \end{aligned}$$

T thus preserves norm. This motivates us to define the following.

Definition 1.1 (Isomorphism of Inner Product Spaces). We say that $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ are isomorphic inner product spaces if there exists a bijective linear map $T : V \rightarrow W$ such that

$$\langle Tx, Ty \rangle_W = \langle x, y \rangle_V \quad \forall x, y \in V.$$

Definition 1.2. $(V, \|\cdot\|_V)$ and $(W, \|\cdot\|_W)$ are said to be isomorphic as Normed Linear Vector Spaces if there exists a bijective linear map $T : V \rightarrow W$ such that T preserves norm i.e. $\|Tv\|_W = \|v\|_V, \forall v \in V$.

Definition 1.3. We have IPVS $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ over a field $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$). Then the metric induced by the inner products say d_V and d_W are defined as

$d_V(x, y) = \|x - y\|_V, \forall x, y \in V$ and $d_W(x, y) = \|x - y\|_W, \forall x, y \in W$. (Here norms are those induced by the inner products)

The metric spaces thus induced are then said to be isomorphic if there exists a bijective linear map $T : V \longrightarrow W$ such that we have $d_W(Tx, Ty) = d_V(x, y)$.

Note that $T : V \longrightarrow \mathbb{F}^n$ as $Tx = (\langle x, u_1 \rangle \ \langle x, u_2 \rangle \ \dots \ \langle x, u_m \rangle)^T$ is a bijective map that preserved inner product and norm. Also,

$$\begin{aligned} \|Tx - Ty\|^2 &= \|T(x - y)\|^2 \\ &= \sum_{i=1}^n |\langle x - y, u_i \rangle|^2 = \|x - y\|^2 \end{aligned}$$

Since, on any finite dimensional space with inner product structure, where metric is induced by norm and norm is induced by inner product ($\langle, \rangle \implies \|\cdot\| \implies d(\cdot, \cdot)$) we have the following result:

Any finite dimensional vector space with inner product structure is isomorphic to $\mathbb{F}^n$ as an *Inner Product Space*, *Normed Vector Space* and as a *Metric Space*.

Hence any n -dimensional inner product space over $\mathbb{F}$ ($\mathbb{R}$ or $\mathbb{C}$) is isomorphic to $\mathbb{F}^n$ as a metric space.

Corollary 1.4. Any inner product space $(V, \langle, \rangle)$ isomorphic to $\mathbb{F}^n$ is complete, i.e. every Cauchy sequence in V converges to a point of V .

Question. Does this also mean that every Inner Product on $\mathbb{R}^n$ is the same?
Answer. Consider $(\mathbb{R}^2, \langle, \rangle_1)$ and $(\mathbb{R}^2, \langle, \rangle_2)$ defined as $\langle (x_1, x_2), (y_1, y_2) \rangle_1 = x_1y_1 + x_2y_2$ and $\langle (u_1, u_2), (v_1, v_2) \rangle_2 = 3u_1v_1 + 2u_2v_2$.

Remark 1.5. If $(V, \langle, \rangle_V)$ is isomorphic to $(W, \langle, \rangle_W)$ as vector spaces need not imply isomorphism as Inner Product Spaces. Consider $f : \mathbb{R} \rightarrow \mathbb{R}$, defined as $f(x) = 2x$ is isomorphism of vector spaces. But f does not preserve Inner Product,

$$\langle f(2), f(5) \rangle = \langle 4, 10 \rangle = 40 \neq 10 = \langle 2, 5 \rangle$$

Remark 1.6. If $V \cong W$ as Inner Product, then $V \cong W$ as Normed Linear Space, further we can conclude that $V \cong W$ as Metric Spaces. Say $T : V \longrightarrow W$ is a bijective linear map which is an isometry, i.e. $\langle Tx, Ty \rangle_W = \langle x, y \rangle_V$ then we have

$$\begin{aligned} \implies \|Tx\|^2 &= \langle Tx, Tx \rangle_W = \langle x, x \rangle_V = \|x\|^2 \\ \implies \|Tx - Ty\|^2 &= \|T(x - y)\|^2 = \|x - y\|^2 \end{aligned}$$

Question. If $V \cong W$ as metric spaces induced from inner products then are $V \cong W$ as inner products as well? *Answer.*

$$\begin{aligned} \|Tx - Ty\|^2 = \|x - y\|^2 &\implies \|Tx\|^2 = \|x\|^2 \quad (\text{Take } y = 0) \\ \langle Tx - Ty, Tx - Ty \rangle & \end{aligned}$$

If $\mathbb{F} = \mathbb{R}$, we have $\langle Tx, Ty \rangle = \langle x, y \rangle$

If $\mathbb{F} = \mathbb{C}$, then $\langle Tx, Ty \rangle = \overline{\langle Ty, Tx \rangle}$ and $\langle Tx, Ty \rangle + \langle Ty, Tx \rangle = \langle x, y \rangle + \langle y, x \rangle = 2a$

Say $\langle Tx, Ty \rangle = a + ib$ then $\langle Tx, Ty \rangle + \langle Ty, Tx \rangle = 2a$. Also if $\langle x, y \rangle = c + id$ then $\langle x, y \rangle + \langle y, x \rangle = 2c$. Thus $a = c$.

$$\begin{aligned} \text{Additionally, } \langle T(ix), Ty \rangle + \langle Ty, T(ix) \rangle &= \langle ix, y \rangle + \langle y, ix \rangle \\ \implies i\langle Tx, Ty \rangle - i\langle Ty, Tx \rangle &= i\langle x, y \rangle - i\langle y, x \rangle \\ \implies i(a + ib) - i(a - ib) &= i(c + id) - i(c - id) \\ \implies b &= d \end{aligned}$$

Hence we have $\langle Tx, Ty \rangle = \langle x, y \rangle$.

Remark 1.7. Alternate Proof of the above uses the polarization, For an isometry T we have $\|Tz\| = \|z\|$ for all z . If $\mathbb{F} = \mathbb{R}$, the real polarization identity

$$\langle u, v \rangle = \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 \right)$$

gives

$$\langle Tx, Ty \rangle = \frac{1}{4} \left(\|Tx + Ty\|^2 - \|Tx - Ty\|^2 \right) = \frac{1}{4} \left(\|x + y\|^2 - \|x - y\|^2 \right) = \langle x, y \rangle.$$

If $\mathbb{F} = \mathbb{C}$, use the complex polarization identity

$$\langle u, v \rangle = \frac{1}{4} \left(\|u + v\|^2 - \|u - v\|^2 + i\|u + iv\|^2 - i\|u - iv\|^2 \right)$$

(with inner product linear in the first variable). Applying T and using $\|Tz\| = \|z\|$ gives $\langle Tx, Ty \rangle = \langle x, y \rangle$.

2 Best Approximation

Aim: To find $\min\{\|x - w\| : w \in W\}$ (and $\arg \min\{\|x - w\| : w \in W\}$) where W is a subset of IPVS V .

Definition 2.1. Let V be an IPVS. $\emptyset \neq W \subset V$. Let $x \in V$. An element $x_0 \in W$ is said to be the best approximation from W to x if

$$\|x - x_0\| \leq \|x - w\|, \quad \forall w \in W$$

i.e. $\|x - x_0\| = \text{dist}(x, W)$

Question. Some natural question arises:

1. Does Best Approximation always exists for a given subset of V and $x \in V$?
2. If it exists, is it unique?
3. Is there any method to find Best Approximation?

Answer: In general best approximation may not exists. Take $V = \mathbb{R}, W = (0, 1) \cap \mathbb{Q}$. Then best approximation does not exist say for $x = 2$.

In general, if best approximation if it exists may not be unique. Consider $V = \mathbb{R}^2, W = \{(x, y) \in \mathbb{R}^2 : |x| = 1 \text{ or } |y| = 1\}$. Then at $x = (0, 0)$, the following are best approximations $(1, 0), (0, 1), (-1, 0), (0, -1)$. For the existence, we have the following result.

Theorem 2.2. *Let V be a finite-dimensional inner product space over $\mathbb{F}$. Let K be a non-empty closed convex subset of V . For every $x \in V$, there exists $x_0 \in K$ such that*

$$\|x - x_0\| = \inf_{w \in K} \|x - w\|.$$

Proof. Let $d = \inf\{\|x - w\| : w \in K\}$. Choose a sequence $\{v_n\}_{n=1}^\infty$ in K such that $\|x - v_n\| \rightarrow d$. Then there exists a sequence $\{v_n\}_{n=1}^\infty$ in K such that $\|x - v_n\| \rightarrow d$. To show $\{v_n\}_{n=1}^\infty$ is Cauchy sequence. Using Pythagoras Theorem, we get,

$$\begin{aligned} \left\| \frac{v_n - v_m}{2} \right\|^2 &= \left\| \frac{v_n - x}{2} - \frac{v_m - x}{2} \right\|^2 \\ &= 2 \left(\left\| \frac{v_n - x}{2} \right\|^2 + \left\| \frac{v_m - x}{2} \right\|^2 \right) - \left\| \frac{v_n - x}{2} + \frac{v_m - x}{2} \right\|^2 \\ &= \frac{1}{2} \left(\|v_n - x\|^2 + \|v_m - x\|^2 \right) - \left\| \frac{v_n + v_m}{2} - x \right\|^2 \\ &\leq \frac{1}{2} \left(\|v_n - x\|^2 + \|v_m - x\|^2 \right) - d^2 \\ &\leq \frac{1}{2} (d^2 + d^2) - d^2 = 0 \quad \text{as } n, m \rightarrow \infty. \end{aligned}$$

Thus $\{v_n\}$ is Cauchy in K . K is a closed subset of a complete metric space (since V is a finite dimensional IPVS). Hence, v_n converges to say $v \in K$.

$$\|x - v\| = \lim_{n \rightarrow \infty} \|x - v_n\| = d = \inf\{\|x - w\| : w \in K\}$$

□

Theorem 2.3. *Let W be a subspace of a finite-dimensional inner product space V . For $x \in V$, an element $x_0 \in W$ is the best approximation to x from W if and only if*

$$x - x_0 \perp W,$$

i.e. $\langle x - x_0, y \rangle = 0$ for all $y \in W$.

Proof. Assume $x - x_0 \perp W$, then for any $y \in W$, $\langle x - x_0, y \rangle = 0$ equivalently $\langle x - x_0, x_0 - y \rangle = 0$, then by using Pythagoras Theorem, we get,

$$\|x - y\|^2 = \|(x - x_0) - (x_0 - y)\|^2 = \|x - x_0\|^2 + \|x_0 - y\|^2 \geq \|x - x_0\|^2.$$

So x_0 is a best approximation.

Conversely, suppose x_0 is a best approximation. Let $z = x - x_0$, $y \in W$, and $t \in \mathbb{F}$. Since W is a subspace, $x_0 + ty \in W$, hence

$$\|z - ty\|^2 = \|x - (x_0 + ty)\|^2 \geq \|x - x_0\|^2 = \|z\|^2.$$

Expanding gives

$$-2 \operatorname{Re}(t \overline{\langle z, y \rangle}) + |t|^2 \|y\|^2 \geq 0.$$

If $\langle z, y \rangle \neq 0$, choose $t = \varepsilon \langle z, y \rangle$ with $\varepsilon > 0$ small. Then the left side becomes

$$-2\varepsilon |\langle z, y \rangle|^2 + \varepsilon^2 |\langle z, y \rangle|^2 \|y\|^2 < 0,$$

a contradiction. Thus $\langle z, y \rangle = 0$. Since $y \in W$ was arbitrary, $x - x_0 \perp W$. $\square$

3 Riesz Representation Theorem

Theorem 3.1 (The Riesz Representation Theorem). *Let $(V, \langle \cdot, \cdot \rangle)$ be a Finite Dimensional IPVS over field $\mathbb{F}$. Let f be linear functional on V , then there exists unique vector $x_0 \in V$ such that,*

$$f(x) = \langle x, x_0 \rangle$$